SLAB	DIMENSIONS	TYPE OF SLAB	TYPE OF PANEL	AREA OF STEEL PROVIDED AT MID SPAN		EXTRA BARS AT SUPPORTS
S2	3100X3300	Two-way slab	Two adjacent edges discontinuous	SHORTER SPAN	8mm dia @ 240mm c/c	8mm dia @ 240mm c/c
				LONGER SPAN	8mm dia @ 210mm c/c	8mm dia @ 210mm c/c
S3	2800X3280	Two-way slab	One shorter edge discontinues	SHORTER SPAN	8mm dia @ 240mm c/c	8mm dia @ 240mm c/c
				LONGER SPAN	8mm dia @ 210mm c/c	8mm dia @ 210mm c/c
S4	2100X3280	Two-way slab	Two adjacent edges discontinuous	SHORTER SPAN	8mm dia @ 240mm c/c	8mm dia @ 240mm c/c
				LONGER SPAN	8mm dia @ 210mm c/c	8mm dia @ 210mm c/c
S5	1920X3280	Two-way slab	Three edges discontinuous	SHORTER SPAN	8mm dia @ 240mm c/c	8mm dia @ 240mm c/c
				LONGER SPAN	8mm dia @ 210mm c/c	8mm dia @ 210mm c/c

Table 4.3: Roof slab details.

Frame	Span	Beam Size (b × D) mm	Eff. Depth(d)mm	M_u KN-m	Ast_{req} mm^2	Min Ast mm^2	Ast Provided mm^2	Main Steel Provided
A2-B2-C2	A2-B2	230 × 300 mm	370	24.75	161.43	144.67	226.20	2 bars of 12mm Ø
	B2-C2	230 × 300 mm	370	32.30	214.16	144.67	226.20	2 bars of 12mm Ø
A3-B3-C3	A3-B3	230 × 300 mm	370	26.48	173.18	144.67	226.20	2 bars of 12mm Ø
	B3-C3	230 × 300 mm	370	33.45	222.44	144.67	226.20	2 bars of 12mm Ø
A1-A2-A3	A1-A2	230 × 300 mm	370	39.15	263.67	144.67	339.29	3 bars of 12mm Ø
	A2-A3	230 × 300 mm	370	48.86	336.90	144.67	339.29	3 bars of 12mm Ø
B1-B2-B3	B1-B2	230 × 300 mm	370	38.78	260.91	144.67	339.29	3 bars of 12mm Ø
	B2-B3	230 × 300 mm	370	27.17	177.94	144.67	226.20	2 bars of 12mm Ø
C1-C2-C3	C1-C2	230 × 300 mm	370	39.54	266.53	144.67	339.29	3 bars of 12mm Ø
	C2-C3	230 × 300 mm	370	37.98	255.22	144.67	339.29	3 bars of 12mm Ø

Table 4.4: Design of Main Reinforcement

Frame	Span	Joint	Factored Shear (Vu) KN	Nom. Shear (τ_v)	Conc. Strength (τ_c) N/mm^2	Shear For Steel (V_{us})	Stirrup Dia & Legs	Spacing (S_v) mm	Max. Spacing	Final Provided Spacing
A2-B2-C2	A2-B2	A2	31.56	0.39	0.30	8.03	8 mm Ø & 2 Leg	2014.36	277.5	275
		B2	37.17	0.44	0.37	5.68	8 mm Ø & 2 Leg	2847.76	277.5	275
	B2-C2	B2	46.13	0.54	0.37	14.64	8 mm Ø & 2 Leg	1104.87	277.5	275
		C2	37.20	0.44	0.30	11.67	8 mm Ø & 2 Leg	1386.06	277.5	275
A3-B3-C3	A3-B3	A3	38.54	0.45	0.30	13.01	8 mm Ø & 2 Leg	1242.91	277.5	275
		B3	37.11	0.44	0.37	5.62	8 mm Ø & 2 Leg	2877.27	277.5	275
	B3-C3	B3	48.02	0.56	0.37	16.53	8 mm Ø & 2 Leg	978.24	277.5	275
		C3	38.30	0.45	0.30	12.77	8 mm Ø & 2 Leg	1266.27	277.5	275
A1-A2-A3	A1-A2	A1	40.98	0.48	0.30	15.45	8 mm Ø & 2 Leg	1046.87	277.5	275
		A2	53.07	0.62	0.43	16.48	8 mm Ø & 2 Leg	981.44	277.5	275
	A2-A3	A2	65.51	0.77	0.43	28.92	8 mm Ø & 2 Leg	559.27	277.5	275
		A3	53.66	0.63	0.30	28.13	8 mm Ø & 2 Leg	574.98	277.5	275
B1-B2-B2	B1-B2	B1	42.21	0.50	0.30	16.68	8 mm Ø & 2 Leg	969.45	277.5	275
		B2	50.94	0.60	0.37	19.45	8 mm Ø & 2 Leg	831.39	277.5	275
	B2-B3	B2	37.77	0.44	0.37	6.28	8 mm Ø & 2 Leg	2574.92	277.5	275
		B3	28.50	0.33	0.30	2.97	8 mm Ø & 2 Leg	5444.62	277.5	275
C1-C2-C3	C1-C2	C1	42.29	0.50	0.30	16.76	8 mm Ø & 2 Leg	964.81	277.5	275
		C2	52.73	0.62	0.37	21.24	8 mm Ø & 2 Leg	761.31	277.5	275
	C2-C3	C2	51.66	0.61	0.37	20.17	8 mm Ø & 2 Leg	801.70	277.5	275
		C3	40.98	0.48	0.30	15.45	8 mm Ø & 2 Leg	1046.62	277.5	275

Table 4.5: Design of Shear Reinforcement (Stirrups

Column	Type	Size (mm)	P_u (KN)	M_{ux} (KN-m)	M_{uy} (KN-m)	Required p (%)	A_{sc} Req (mm^2)	Main Steel Provided	A_{sc} Prov (mm^2)	Ties Provided
C1	Bi-axial	230 x 300	53.94	9.73	8.04	0.8	552	4 bars of 16 mm Ø	804	8mm Ø @ 200mm c/c
A2	Uni-axial	230 x 300	109.43	2.48	2.19	0.8	552	4 bars of 16 mm Ø	804	8mm Ø @ 230mm c/c
B2	Bi-axial	230 x 300	114.67	2.97	3.58	0.8	552	4 bars of 16 mm Ø	804	8mm Ø @ 200mm c/c
C2	Uni-axial	230 x 300	83.94	1.68	7.31	0.8	552	4 bars of 16 mm Ø	804	8mm Ø @ 230mm c/c
A3	Bi-axial	230 x 300	76.20	12.18	1.52	0.8	552	4 bars of 16 mm Ø	804	8mm Ø @ 200mm c/c
B3	Uni-axial	230 x 300	75.75	6.32	4.88	0.8	552	4 bars of 16 mm Ø	804	8mm Ø @ 230mm c/c
C3	Bi-axial	230 x 300	52.85	9.22	7.17	0.8	552	4 bars of 16 mm Ø	804	8mm Ø @ 200mm c/c

Table 4.6: Design of Columns C1, A2, B2, C2, A3, B3 & C3

Footing	Factored Load P_u	Upward Pressure q_u ($KN./m^2$)	Max Moment M_u (KN-m)	One-Way Shear V_{u1} (KN)	Punching Shear V_{u2} (KN)	Status
B1	133.04	133.04	9.86	25.28	105.10	SAFE
C1	53.94	53.94	4.00	10.25	42.61	SAFE
A2	109.43	109.43	8.11	20.79	86.45	SAFE
B2	114.67	114.67	8.50	21.79	90.59	SAFE
C2	83.94	83.94	6.22	15.95	66.31	SAFE
A3	76.20	76.20	5.65	14.48	60.20	SAFE
B3	75.75	75.75	5.61	14.39	59.84	SAFE
C3	52.85	52.85	3.92	10.04	41.75	SAFE

Table 4.7: Design of footing B1, C1, A2, B2, C2, A3, B3 & C3